

Router IP Address & DHCP

Assignment Answers

Important: Router settings and IP addresses are specific to the local network. The example values below show what a completed answer can look like, but they should be replaced with the actual values from the student's own router/device.

1. Router admin-panel login page

To find the router's admin-panel address on Windows, connect the laptop to the home Wi-Fi and open **Command Prompt**. Type **ipconfig** and press Enter. Find the Wi-Fi adapter and look for **Default Gateway**. Common addresses are 192.168.0.1 or 192.168.1.1.

Step	Action
1	Connect the laptop to the home Wi-Fi network.
2	Open Command Prompt and run: ipconfig
3	Find the Default Gateway under the Wi-Fi adapter.
4	Open a web browser and enter the gateway address, for example http://192.168.1.1
5	Enter the router administrator username/password when prompted.

Screenshot to include: Take a screenshot of the router login page and paste it in the box below.

PASTE ROUTER LOGIN PAGE SCREENSHOT HERE

Do not include or share the router administrator password.

Example: If ipconfig shows Default Gateway = 192.168.1.1, the router login page may be opened at 192.168.1.1.

2. Current IP address, subnet mask and default gateway

On Windows, run **ipconfig** and locate the wireless/Wi-Fi adapter. Record the IPv4 Address, Subnet Mask and Default Gateway. On a phone, open the connected Wi-Fi network's details and look for IP address, subnet mask/prefix length and router/gateway.

Network setting	Example answer	Actual value to record
IPv4 address	192.168.1.23	_____
Subnet mask	255.255.255.0	_____
Default gateway	192.168.1.1	_____

Example completed answer: IPv4 address = 192.168.1.23; subnet mask = 255.255.255.0; default gateway = 192.168.1.1. These are example values only and are not the user's actual network values.

3. Change DHCP to a static IP, test Internet access, then revert

Example setup: Suppose the router is 192.168.1.1 and the subnet mask is 255.255.255.0. A suitable unused static address could be 192.168.1.50. The exact address must be chosen so that it is not already being used by another device and does not conflict with the router's DHCP pool.

Windows 10/11 procedure

1	Open Settings → Network & Internet → Wi-Fi → Hardware properties / adapter options.
2	Open the Wi-Fi adapter's IPv4 properties. Select Internet Protocol Version 4 (TCP/IPv4).
3	Change from "Obtain an IP address automatically" to "Use the following IP address."
4	Example: IP address 192.168.1.50; Subnet mask 255.255.255.0; Default gateway 192.168.1.1.
5	For DNS, use the router address (192.168.1.1) or the DNS server specified by the network administrator.
6	Save the settings. Open a browser and test a website. You can also run ping 192.168.1.1 and ping 8.8.8.8.
7	Record whether the Internet worked with the static address.
8	IMPORTANT: Return the IPv4 setting to "Obtain an IP address automatically" and DNS to automatic after testing.

Example test result: "Internet access worked after assigning 192.168.1.50 because the address was in the same subnet as the router, the gateway was correct, and the address was not already in use." If the test fails, return to DHCP and check the IP, mask, gateway, DNS and possible address conflicts.

4. DHCP range and duplicate IP addresses

Log in to the router and open the LAN/DHCP settings page. Look for fields such as **DHCP Server**, **Starting IP Address**, **Ending IP Address** or **DHCP Address Pool**.

Item	Example	Actual router value
DHCP starting address	192.168.1.100	_____
DHCP ending address	192.168.1.200	_____
DHCP range	192.168.1.100 – 192.168.1.200	_____

What happens if two devices receive the same IP address? An IP address conflict occurs. The two devices may have intermittent or no network/Internet connectivity because traffic intended for one device can be associated with the wrong device. Applications may disconnect, and Windows or the router may show a duplicate-address/conflict warning.

Important: In a normal DHCP network, the DHCP server tracks leased addresses to reduce the chance of duplicates. Conflicts can still happen if someone manually assigns a static address inside the DHCP pool or if a device is configured incorrectly.

Final checklist

- Router login page screenshot added
- Actual IPv4 address recorded
- Actual subnet mask recorded
- Actual default gateway recorded
- Static IP tested
- Device returned to DHCP/automatic settings
- Actual DHCP range recorded
- Duplicate-IP effect explained